



East West University

Task 3 Report

Course Code: CSE475

Course Name: Machine Learning

Section: 2

Group	Group04
Dataset	BloodMNIST (MedMNIST v2)
Track	Track 2 + Track 3 (GCN + CNN+Att)
Proposed Direction	CNN + Attention + GCN

Submitted To,

Dr Raihan Ul Islam

Associate Professor

Department of Computer Science & Engineering

East West University

Submitted By,

Member Name	Student ID
Fahmida Jannat	2025-1-60-334
Nishat Subha Mithela	2023-1-60-248
Nusrat Jahan	2022-2-60-128
Humayara Nahar Moumita	2021-2-60-121

1. Introduction

This report presents Task 3 of the BloodMNIST classification project. The main objectives of Task 3 were to improve the model developed in Task 2, determine the contribution of important architectural components through ablation experiments, compare the improved model with the strongest CNN baseline, evaluate stability using five-fold cross-validation and statistical testing, and explain individual predictions using Grad-CAM, LIME, and graph-node importance.

In Task 2, the project established MobileNetV3Large as the strongest CNN baseline, while the first proposed graph-based model used a frozen ImageNet-pretrained ResNet50 backbone, channel-and-spatial attention, a 7×7 spatial feature map represented as 49 graph nodes, and a Residual GCN. The Task 2 report recorded 91.26% accuracy and 90.38% macro-F1 for MobileNetV3Large and 91.70% accuracy and 90.87% macro-F1 for the proposed GCN model. The Task 2 graph construction used 49 nodes and a four-neighbor spatial grid.

Task 3 continued from this model and performed controlled experiments involving GCN removal, attention removal, graph connectivity, GCN depth, dropout, and GNN variant. The final model was then evaluated on the held-out test set and compared with a reproduced MobileNetV3Large baseline. The project brief also requires five-fold cross-validation with mean $\pm$ standard deviation and a statistically appropriate significance test.

2. Experimental Setup

2.1 Dataset and Official Splits

BloodMNIST from MedMNIST v2 was used throughout Task 3.

The official dataset contains:

- Training images: 11,959
- Validation images: 1,712
- Test images: 3,421
- Image size: $28 \times 28 \times 3$
- Number of classes: 8

The official train, validation, and test splits were preserved for the main experiments. Task 3 reproduced the same preprocessing pipeline used in Task 2 rather than creating a new random train/test partition.

2.2 Preprocessing

The Task 2 preprocessing pipeline was reproduced exactly for the Task 3 experiments.

The main settings were:

- Input resolution: $224 \times 224 \times 3$
- Batch size: 32
- Training-only undersampling
- Balanced class weights for the GCN training
- Adam optimizer
- Initial learning rate: 0.001
- ReduceLROnPlateau learning-rate scheduling
- Early stopping based on validation loss
- Random seed: 42

The original training set contained 11,959 images. After the same Task 2 undersampling procedure, 8,207 training samples were used. The validation and test sets remained unchanged at 1,712 and 3,421 images respectively.

2.3 Original Task 2 GCN Reproduction

Before starting the ablation study, the original Task 2 GCN was rebuilt and trained using the same configuration.

The original architecture used:

- Frozen ImageNet-pretrained ResNet50
- 7×7 feature map
- 49 graph nodes
- Four-neighbor spatial graph
- Self-loops
- Normalized adjacency
- Channel and spatial attention
- Three Residual GCN layers
- GCN dropout = 0.20
- Classifier dropout = 0.35

The reproduced original GCN achieved the following on the held-out test set:

Metric	Original Task 2 GCN
Accuracy	91.61%
Macro Precision	91.12%
Macro Recall	90.26%
Macro-F1	90.59%

Weighted-F1	91.66%
Macro ROC-AUC	99.23%
Weighted ROC-AUC	99.21%
Cohen's Kappa	0.9019
MCC	0.9022
Log Loss	0.3099

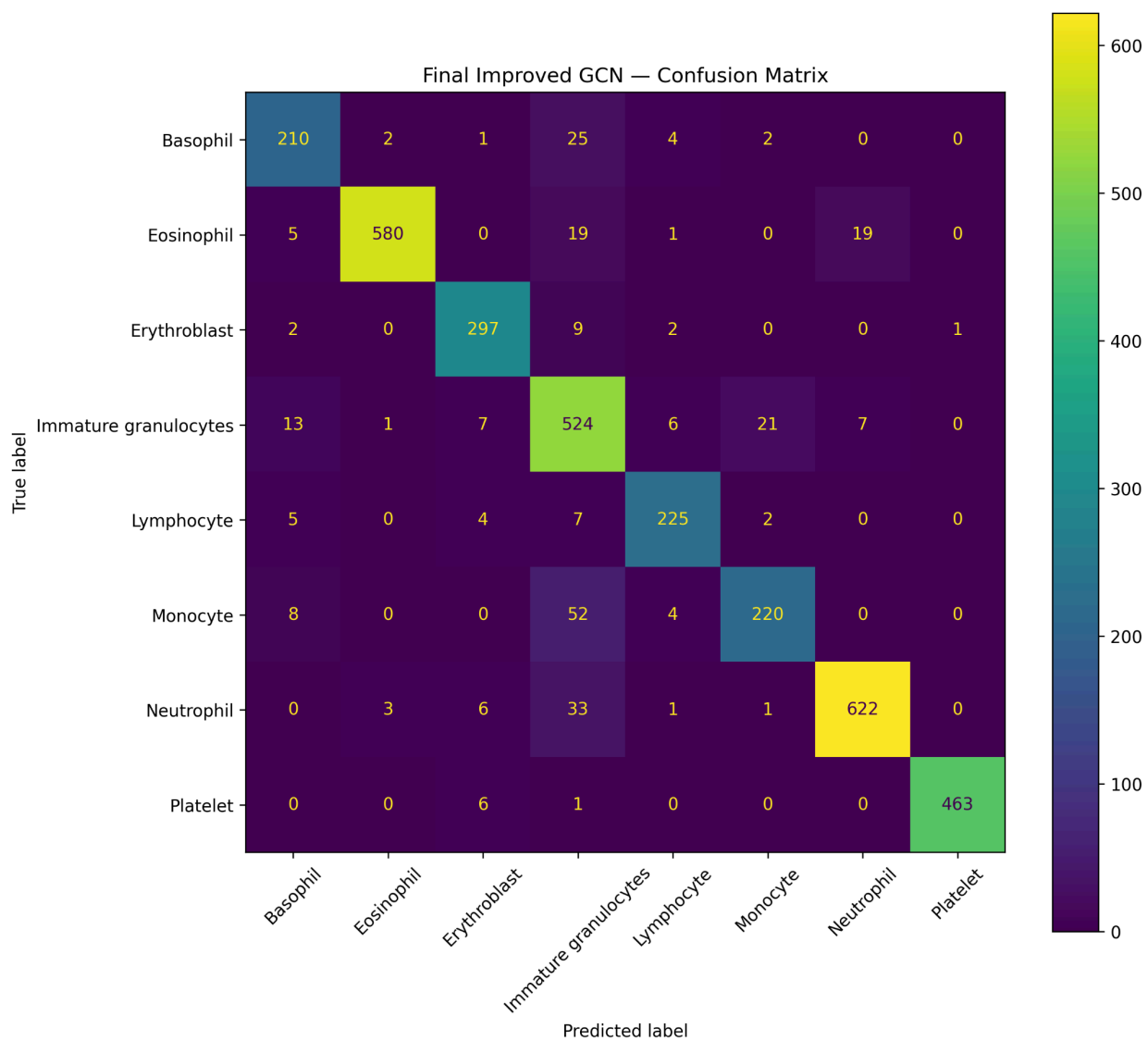


Figure 1. Original Task 2 ResNet50 + Attention + Residual GCN configuration used as the Task 3 starting point.

3. Task 3 Ablation Study

The Task 3 ablation study investigated whether the main components of the proposed architecture actually contributed to validation performance.

The experiments were based on validation data. The held-out test set was reserved for the final evaluation after the final configuration was locked.

The project specification emphasizes macro-F1 and class-wise recall for imbalanced classification, so macro-F1 was used as the primary comparison metric in the ablation experiments.

4. Ablation A0 — Removing the GCN

The first experiment removed the GCN while retaining the frozen ResNet50 backbone and the channel-and-spatial attention module.

This created a directly comparable CNN model using the same backbone and attention representation but without graph-based relational reasoning.

Experiment	Accuracy	Macro Precision	Macro Recall	Macro-F1	Macro ROC-AUC
Original ResNet50 + Attention + Residual GCN	91.18%	90.07%	90.06%	89.97%	99.31%
ResNet50 + Attention, no GCN	90.54%	89.25%	89.57%	89.36%	99.20%

The GCN increased validation accuracy by 0.64 percentage points and macro-F1 by 0.61 percentage points compared with the otherwise comparable CNN model.

This result provides evidence that graph-based relational processing contributed useful information beyond the CNN and attention representation.

Experiment	Model	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss	Parameters
Original GCN	ResNet50 + Attention + Residual GCN	0.9118	0.9007	0.9006	0.8997	0.9117	0.9931	0.3001	25505580
A0 - No GCN	ResNet50 + Attention CNN	0.9054	0.8925	0.8957	0.8936	0.9044	0.9920	0.2819	24742153

Figure 2. Validation comparison between the original GCN model and the same CNN-attention pipeline without the GCN

5. Ablation A1 — Removing Attention

The second experiment removed the attention module while retaining the ResNet50 backbone and Residual GCN.

Experiment	Attention	Accuracy	Macro Precision	Macro Recall	Macro-F1	Macro ROC-AUC
Original GCN	Channel + Spatial	91.18%	90.07%	90.06%	89.97%	99.31%
A1 — No Attention	None	91.30%	89.78%	90.91%	90.18%	99.36%

Removing attention slightly increased validation accuracy by 0.12 percentage points and macro-F1 by 0.21 percentage points.

The notebook therefore recorded the decision to remove attention from the subsequent improvement pipeline.

This is an important finding because it shows that the attention block used in Task 2 was not automatically beneficial under the Task 3 experimental setup.

Experiment	Model	Attention	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss
A1 - No Attention	ResNet50 + Residual GCN	None	0.9130	0.8978	0.9091	0.9018	0.9124	0.9936	0.2894
Original GCN	ResNet50 + Attention + Residual GCN	Channel + Spatial	0.9118	0.9007	0.9006	0.8997	0.9117	0.9931	0.3001

Figure 3. Effect of removing the channel-and-spatial attention module.

6. Ablation A2 — Four-Neighbor versus Eight-Neighbor Graph

The third experiment investigated graph connectivity.

The original four-neighbor graph connects each node to its horizontal and vertical spatial neighbors. The eight-neighbor version additionally connects diagonal neighbors.

The isolated graph comparison produced:

Graph	Accuracy	Macro Precision	Macro Recall	Macro-F1	Macro ROC-AUC
Four-neighbor	91.30%	89.78%	90.91%	90.18%	99.36%
Eight-neighbor	91.71%	91.52%	90.36%	90.88%	99.36%

The eight-neighbor graph achieved a higher validation macro-F1 by 0.70 percentage points in this isolated comparison.

However, the subsequent Task 3 pipeline retained the four-neighbor graph for the final locked configuration. Therefore, the eight-neighbor result is reported as an ablation finding, not as the final graph configuration.

Experiment	Graph	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss	Parameters
A2	8-neighbour	0.9171	0.9152	0.9036	0.9088	0.9176	0.9936	0.2624	24878763
A1	4-neighbour	0.9130	0.8978	0.9091	0.9018	0.9124	0.9936	0.2894	24878763

Figure 4. Four-neighbor and eight-neighbor graph connectivity comparison.

7. GCN Depth Ablation

The number of GCN layers was then varied between one, two, and three layers.

The executed validation experiment produced:

GCN Depth	Accuracy	Macro Precision	Macro Recall	Macro-F1	Macro ROC-AUC
1 layer	89.95%	88.96%	89.65%	88.98%	99.26%
2 layers	91.59%	90.52%	90.96%	90.61%	99.36%
3 layers	91.30%	89.78%	90.91%	90.18%	99.36%

The two-layer configuration achieved the highest validation macro-F1 among the explicitly compared depth settings, at 90.61%.

The one-layer model performed substantially worse, while the three-layer model did not exceed the two-layer result in this isolated depth experiment.

The final consolidated Task 3 pipeline nevertheless used three GCN layers. Therefore, the depth experiment is presented as evidence that GCN depth materially affects performance rather than as a claim that three layers were the best isolated depth result.

GCN_Depth	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss	Parameters	Training_
2	0.9159	0.9052	0.9096	0.9061	0.9155	0.9936	0.2956	24744266	7.3962
3	0.9130	0.8978	0.9091	0.9018	0.9124	0.9936	0.2894	24878763	7.9751
1	0.8995	0.8896	0.8965	0.8898	0.9003	0.9926	0.2956	24151017	6.2771

Figure 5. Effect of GCN depth on validation performance.

8. GCN Dropout Ablation

GCN dropout was then varied.

The final consolidated ablation results were:

GCN Dropout	Accuracy	Macro-F1	Macro-AUC
0.1	91.76%	90.97%	99.32%
0.2	91.94%	90.83%	99.42%
0.3	92.64%	91.87%	99.41%

The dropout rate of 0.30 produced the highest validation macro-F1 in the final consolidated ablation summary and was therefore used in the locked final configuration.

GCN_Dropout	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss	Parameters	Training_
0.3	0.9223	0.9194	0.9139	0.9160	0.9229	0.9938	0.2683	24878763	7.0252
0.1	0.9147	0.9050	0.9110	0.9076	0.9145	0.9939	0.2470	24878763	5.6676
0.2	0.9130	0.8978	0.9091	0.9018	0.9124	0.9936	0.2894	24878763	7.9751

Figure 6. Validation performance for different GCN dropout rates.

9. GNN Variant Ablation — Residual GCN versus GraphSAGE

The GNN variant was also changed from Residual GCN to GraphSAGE using mean aggregation.

GNN Variant	Accuracy	Macro Precision	Macro Recall	Macro-F1	Macro ROC-AUC
Residual GCN	92.23%	91.94%	91.39%	91.60%	99.38%
GraphSAGE	90.60%	90.16%	89.24%	89.60%	99.23%

Residual GCN achieved a 2.00 percentage-point advantage in validation macro-F1 over GraphSAGE.

The experiment therefore supports retaining Residual GCN as the graph-learning method in the final model.

Variant	Accuracy	Macro_Precision	Macro_Recall	Macro_F1	Weighted_F1	Macro_ROC_AUC	Log_Loss	Parameters	Training
Residual GCN	0.9223	0.9194	0.9139	0.916	0.9229	0.9938	0.2683	24878763	7.0252
GraphSAGE	0.9060	0.9016	0.8924	0.896	0.9058	0.9923	0.2831	24878763	5.8194

Figure 7. Validation comparison between Residual GCN and GraphSAGE.

10. Final Consolidated Ablation Result

The final Task 3 notebook consolidated the tested configurations as follows:

Experiment	Change	Attention	Graph	Depth	Dropout	Variant	Accuracy	Macro-F1	Macro-AUC
Dropout-0.30	GCN dropout = 0.30	None	4-neighbo r	3	0.3	Residua l GCN	92.64%	91.87%	99.41%
Dropout-0.10	GCN dropout = 0.10	None	4-neighbo r	3	0.1	Residua l GCN	91.76%	90.97%	99.32%
A1	Remove Attention	None	4-neighbo r	3	0.2	Residua l GCN	91.94%	90.83%	99.42%

A2	4 $\rightarrow$ 8 neighbor	None	8-neighbor	3	0.2	Residual GCN	91.82%	90.83%	99.37%
GraphSAGE	GCN $\rightarrow$ GraphSAGE	None	4-neighbor	3	0.3	GraphSAGE	91.59%	90.52%	99.40%
Depth-2	GCN depth = 2	None	4-neighbor	2	0.2	Residual GCN	91.06%	90.13%	99.34%
Depth-1	GCN depth = 1	None	4-neighbor	1	0.2	Residual GCN	91.24%	90.02%	99.35%
Original	Original Task 2 model	Channel + Spatial	4-neighbor	3	0.2	Residual GCN	90.77%	89.62%	99.31%
A0	Remove GCN	Channel + Spatial	None	0	—	CNN only	90.30%	89.20%	99.15%

The final consolidated notebook reports an improvement of:

- **+1.87 percentage points** in validation accuracy over the original Task 2 GCN;
- **+2.25 percentage points** in validation macro-F1;
- **+0.10 percentage points** in validation macro-AUC.

The final selected configuration was therefore recorded as:

Component	Final Configuration
Backbone	ImageNet-pretrained ResNet50
Backbone trainable	No
Input size	$224 \times 224 \times 3$
Attention	None
Feature map	$7 \times 7 \times 2048$
Graph nodes	49
Graph arrangement	7×7
Graph connectivity	Four-neighbor
Self-loops	Yes

Adjacency normalization	$D^{-1/2} A D^{-1/2}$
GNN variant	Residual GCN
GCN depth	3
Hidden dimensions	$256 \rightarrow 256 \rightarrow 128$
GCN dropout	0.3
Residual connection	Yes
Graph pooling	Global Average Pooling
Classifier dense layer	256
Classifier dropout	0.35
Optimizer	Adam
Initial learning rate	0.001
LR scheduler	ReduceLROnPlateau
Class weighting	Balanced
Selection metric	Validation Macro-F1

The locked configuration achieved:

- Validation accuracy: **92.64%**
- Validation macro-F1: **91.87%**
- Validation macro-AUC: **99.41%**

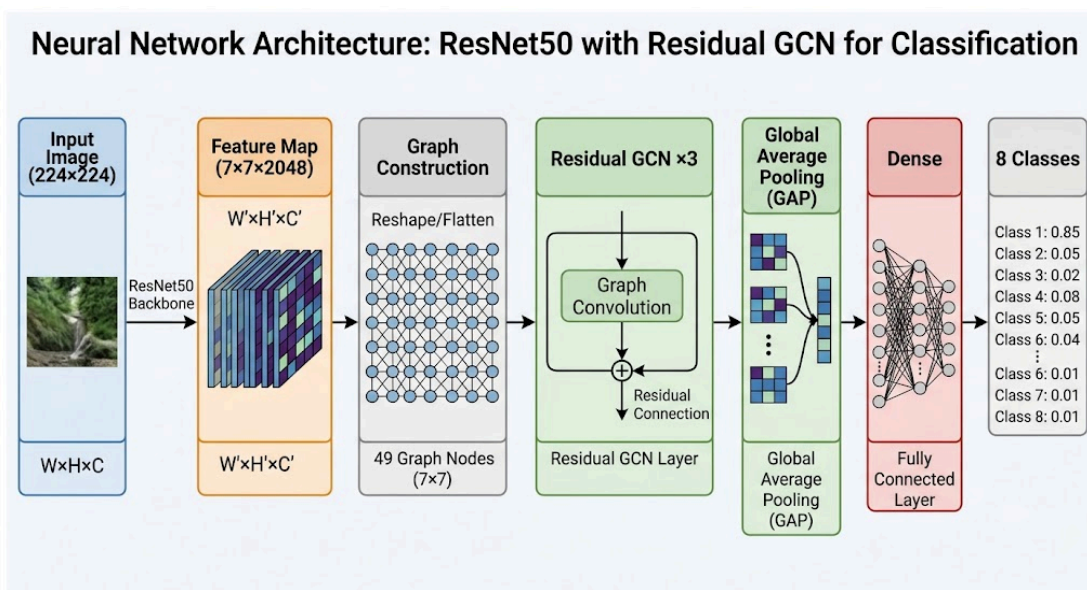


Figure 8. Final locked Task 3 ResNet50 + Residual GCN architecture

11. Held-Out Test Evaluation

After the architecture was locked using validation data, the final model was evaluated on the untouched test set.

The final model produced:

Metric	Final Residual GCN
Accuracy	91.82%
Macro Precision	91.74%
Macro Recall	90.87%
Macro-F1	91.18%
Weighted-F1	91.90%
Macro ROC-AUC	99.33%
Weighted ROC-AUC	99.33%
Cohen's Kappa	0.9043
MCC	0.9049
Log Loss	0.2764
Inference time	17.741 s
Inference time / image	5.186 ms

The final model outperformed the reproduced original Task 2 GCN by:

- Accuracy: **+0.20 percentage points**
- Macro-F1: **+0.59 percentage points**

The per-class results were:

Class	Precision	Recall	F1
--------------	------------------	---------------	-----------

Basophil	86.42%	86.07%	86.24%
Eosinophil	98.98%	92.95%	95.87%
Erythroblast	92.52%	95.50%	93.99%
Immature granulocytes	78.21%	90.50%	83.91%
Lymphocyte	92.59%	92.59%	92.59%
Monocyte	89.43%	77.46%	83.02%
Neutrophil	95.99%	93.39%	94.67%
Platelet	99.78%	98.51%	99.14%

The results show especially strong performance for Platelet, Eosinophil, Neutrophil and Erythroblast, while Immature granulocytes and Monocyte remain relatively more difficult classes.

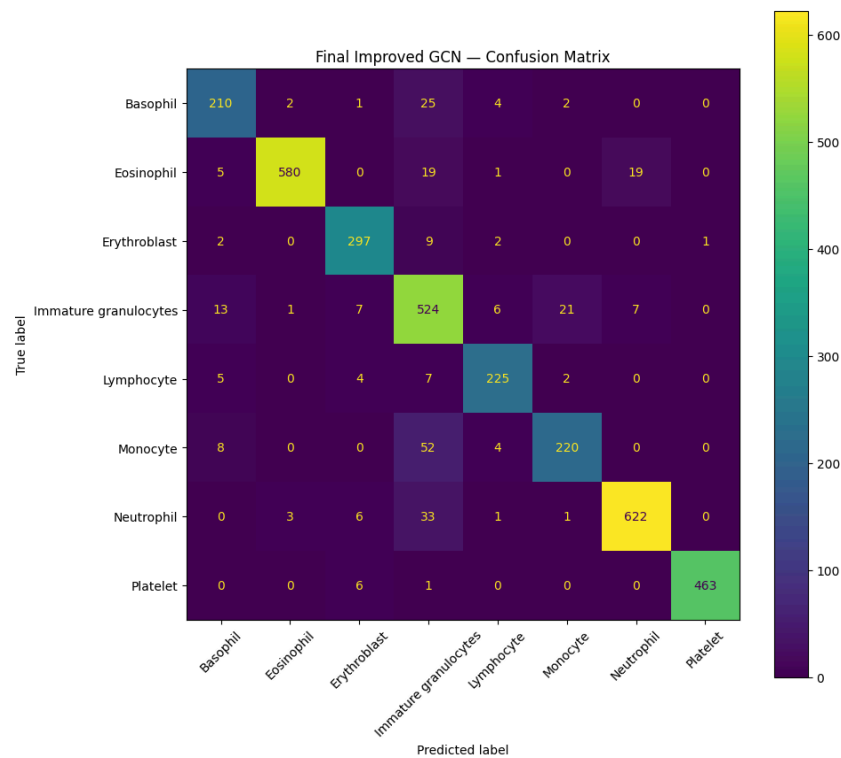


Figure 9. Confusion matrix of the final Residual GCN on the held-out BloodMNIST test set.

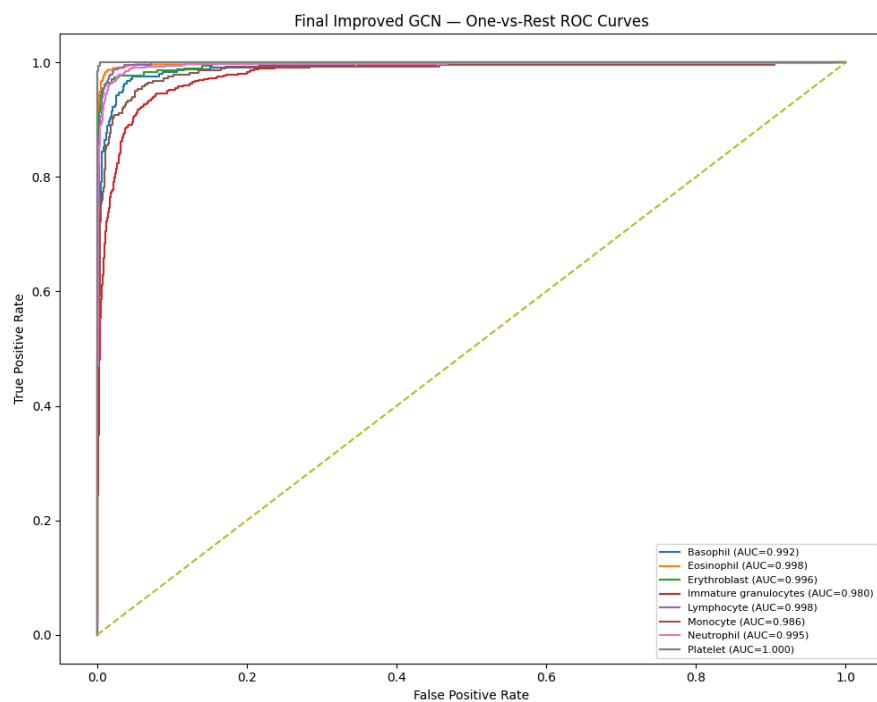


Figure 10. One-vs-rest ROC curves for the final Residual GCN across the eight BloodMNIST classes.

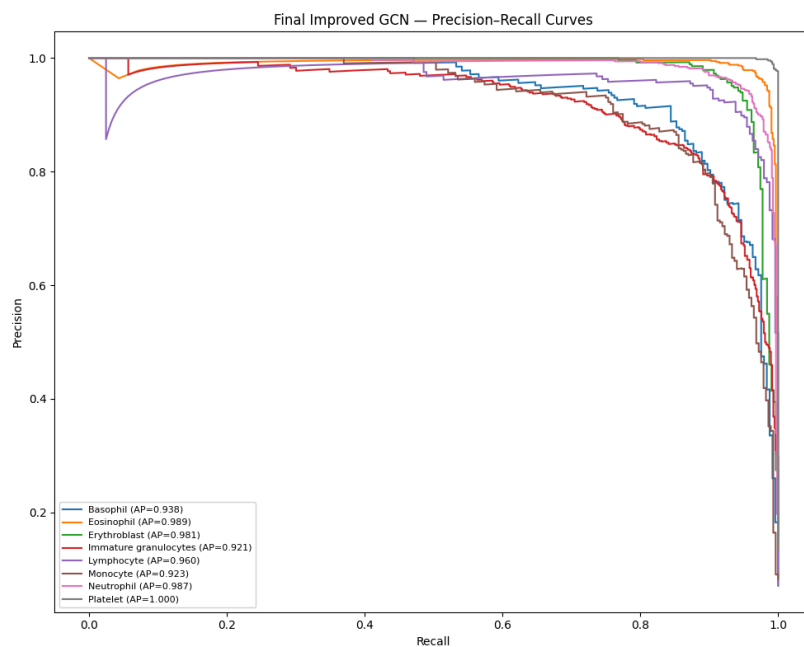


Figure 11. Precision-recall curves for the final Residual GCN.

12. Comparison with the CNN Baseline

The strongest Task 2 CNN baseline was MobileNetV3Large.

The Task 3 notebook reproduced the MobileNetV3Large model using the same Task 3 data pipeline. The reproduction produced:

Metric	Task 2 Reported	Task 3 Reproduced
Accuracy	91.26%	91.08%
Macro Precision	90.42%	89.98%
Macro Recall	90.42%	90.68%
Macro-F1	90.38%	90.28%
Weighted-F1	—	91.12%
Macro ROC-AUC	—	99.26%
Log Loss	—	0.2638

The difference between the Task 2 reported and Task 3 reproduced baseline was small:

- Accuracy difference: **−0.18 percentage points**
- Macro-F1 difference: **−0.10 percentage points**

For the final Task 3 comparison, the notebook also compared the final model with the original Task 2 reported MobileNetV3Large result:

Model	Accuracy	Macro Precision	Macro Recall	Macro-F1
MobileNetV3Large	91.26%	90.42%	90.42%	90.38%
Original ResNet50 + Attention + GCN	91.61%	91.12%	90.26%	90.59%
Final ResNet50 + Residual GCN	91.82%	91.74%	90.87%	91.18%

Relative to the Task 2 reported MobileNetV3Large result, the final GCN improved:

- Accuracy by **+0.56 percentage points**
- Macro-F1 by **+0.80 percentage points**

Relative to the reproduced Task 3 MobileNetV3Large result, the improvement was:

- Accuracy: **+0.73 percentage points**
- Macro-F1: **+0.89 percentage points**

The reproduced baseline is useful for interpreting the Task 3 statistics because it was evaluated under the same Task 3 pipeline.

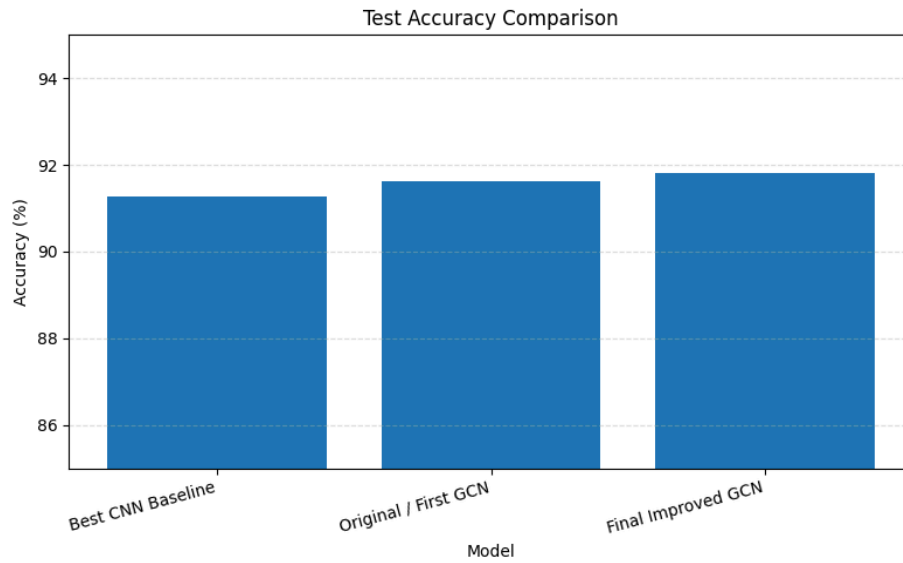


Figure 12. Held-out test accuracy comparison between MobileNetV3Large, the original GCN, and the final improved GCN.

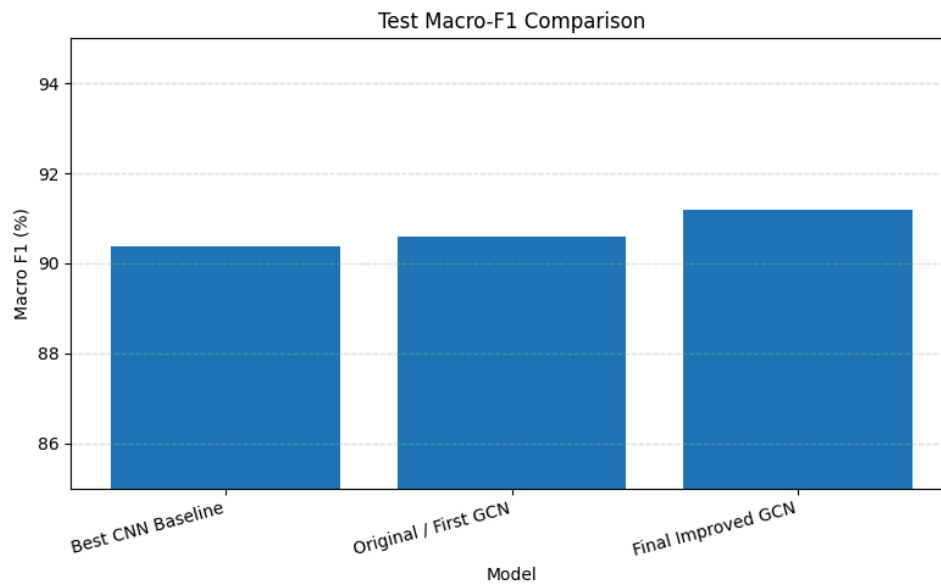


Figure 13. Held-out test macro-F1 comparison between MobileNetV3Large, the original GCN, and the final improved GCN.

13. Five-Fold Cross-Validation

To examine whether the improvement was stable across different development subsets, five-fold cross-validation was performed.

The official training and validation sets were combined into a development set of 13,671 samples.

Before assigning folds, exact image hashes were computed.

The duplicate audit found:

- Development samples: 13,671
- Unique image hashes: 13,660
- Duplicate instances: 11
- Cross-label duplicate groups: 0

The resulting folds had zero duplicate-group overlap.

Fold	Training Samples	Validation Samples	Duplicate-Group Overlap
1	10,939	2,732	0
2	10,936	2,735	0
3	10,938	2,733	0
4	10,938	2,733	0
5	10,933	2,738	0

The held-out test set was not used in the cross-validation procedure.

The five-fold results were:

Model	Accuracy	Macro-F1	Macro ROC-AUC	Log Loss
MobileNetV3Large	90.61% $\pm$ 0.16%	89.34% $\pm$ 0.39%	99.25% $\pm$ 0.06%	0.2688 $\pm$ 0.0082
Final Residual GCN	91.70% $\pm$ 0.57%	90.62% $\pm$ 0.70%	99.37% $\pm$ 0.09%	0.2712 $\pm$ 0.0186

The fold-wise macro-F1 results were:

Fold	MobileNetV3Large	Final GCN	Gain
1	88.84%	90.31%	1.47
2	89.59%	91.74%	2.15
3	89.17%	90.83%	1.65
4	89.84%	90.22%	0.38
5	89.26%	89.99%	0.73

The final GCN achieved a higher macro-F1 in all five folds, with an average improvement of **+1.28 percentage points**.

MobileNetV3Large Macro-F1 : 89.34% $\pm$ 0.39%
Final Residual GCN Macro-F1: 90.62% $\pm$ 0.70%
Mean Macro-F1 gain : +1.28 pp

Figure 14. Five-fold Macro-F1 comparison between MobileNetV3Large and the final Residual GCN.

14. Statistical Significance Testing

14.1 McNemar Test

McNemar's exact test was applied to the paired held-out predictions of the reproduced MobileNetV3Large baseline and the final Residual GCN.

The contingency table was:

	Final GCN Correct	Final GCN Wrong
Baseline Correct	2,952	164
Baseline Wrong	189	116

Therefore:

- Baseline correct, final wrong: 164
- Baseline wrong, final correct: 189
- Discordant pairs: 353
- Corrected chi-square: 1.6317
- Exact p-value: **0.201397**
- $\alpha = 0.05$

The result is **not statistically significant** at the 0.05 level.

Although the final GCN corrected more baseline errors than the baseline corrected final-GCN errors, the evidence is insufficient to reject the null hypothesis of equal prediction behavior.

This interpretation is consistent with the project brief, which states that McNemar's test should be used when two models are evaluated on the same held-out samples.

```
----
McNemar exact p-value : 0.201397
McNemar result       : Not significant
```

Figure 15. Paired prediction outcomes used for the McNemar test.

14.2 Wilcoxon Signed-Rank Test

A two-sided Wilcoxon signed-rank test was applied to the five paired cross-validation Macro-F1 values.

The fold-wise gains were:

- +1.472 percentage points
- +2.151 percentage points
- +1.653 percentage points
- +0.379 percentage points
- +0.730 percentage points

The test produced:

- Wilcoxon statistic: **0.0000**
- p-value: **0.062500**
- $\alpha = 0.05$

The result is therefore **not statistically significant** at the 0.05 level.

```
Wilcoxon p-value      : 0.062500
Wilcoxon result      : Not significant
```

Because all five fold differences favored the final GCN, the test demonstrates consistency of the improvement in the observed folds. However, with only five paired observations, the exact two-sided Wilcoxon test cannot reach $p < 0.05$ in this all-positive-difference situation.

The appropriate conclusion is therefore that the final GCN shows a **consistent observed improvement**, but the available five-fold statistical test does not establish significance at $\alpha = 0.05$.

15. Explainability Analysis

The Task 3 explainability notebook analyzed two test examples using three methods:

1. Grad-CAM
2. LIME
3. GCN node importance

The selected examples were:

Example	Test Index	True Class	Predicted Class	Confidence
Correct	1024	Eosinophil	Eosinophil	1
Misclassified	3126	Neutrophil	Eosinophil	1

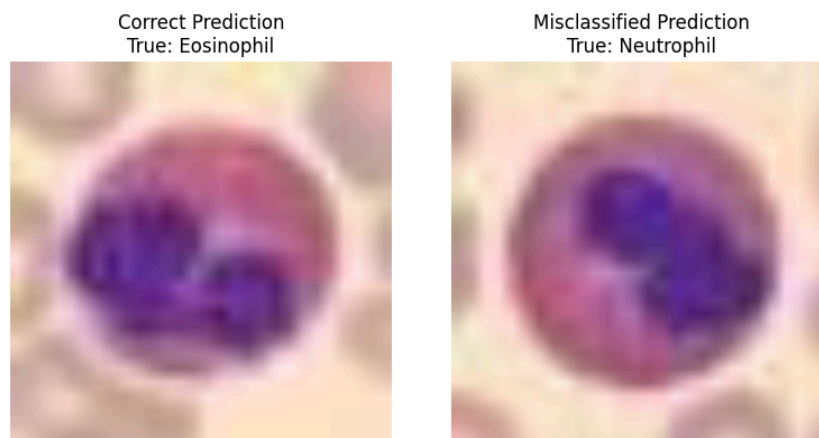


Figure 16. Correctly classified and misclassified examples selected for explainability analysis.

16. Grad-CAM Analysis

Grad-CAM was applied to the ResNet50 feature representation to highlight spatial regions that influenced the final prediction.

For the correctly classified Eosinophil example, Grad-CAM identifies the image regions that contributed to the predicted Eosinophil class.

For the misclassified Neutrophil example, Grad-CAM identifies the regions that contributed to the incorrect Eosinophil prediction.

The purpose of the comparison is to examine how the model's spatial evidence differs between a successful and an unsuccessful prediction.

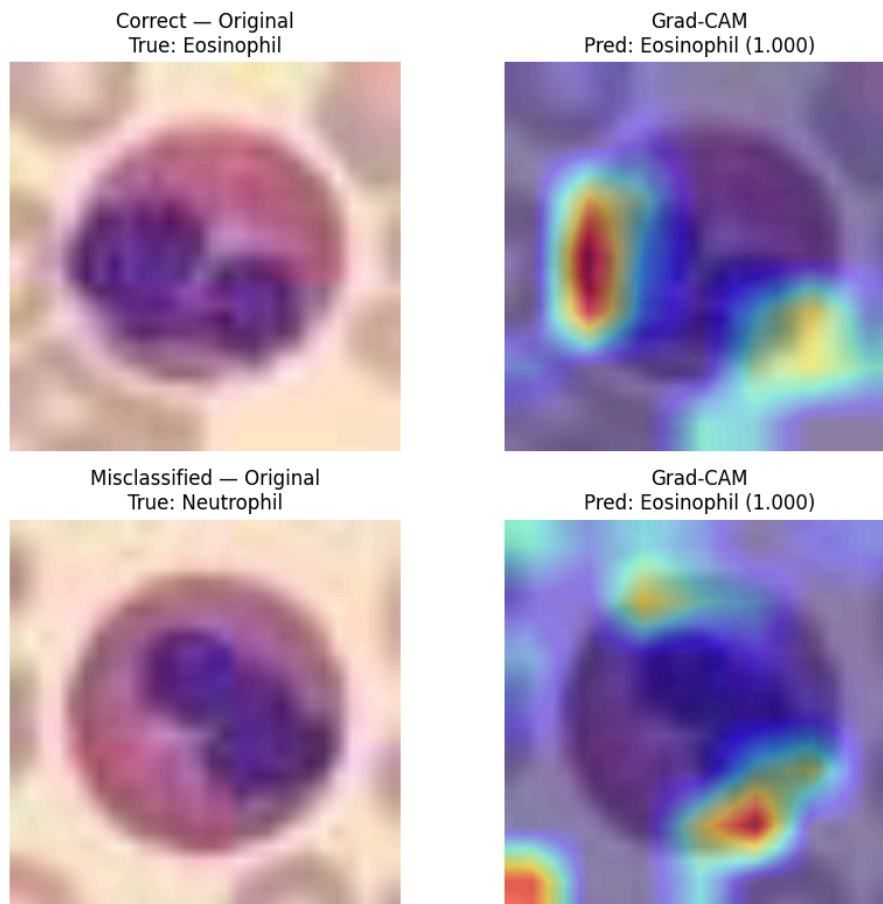


Figure 17. Grad-CAM explanations for the correct and misclassified test examples.

17. LIME Analysis

LIME was applied using image superpixels and local perturbations.

For each selected example, LIME identifies image regions whose presence or absence changes the local prediction behavior.

The correct example provides an explanation of the model's Eosinophil decision, while the misclassified example shows which image regions supported the incorrect Eosinophil prediction.

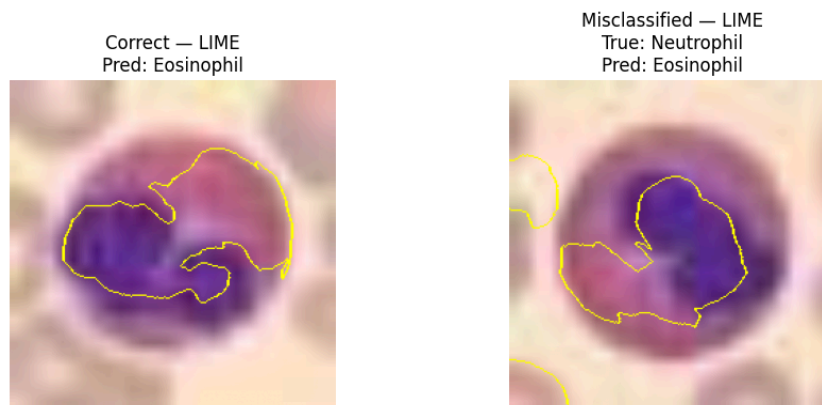


Figure 18. LIME explanations for the correct and misclassified predictions.

18. GCN Node-Importance Analysis

The final model converts the 7×7 ResNet50 feature map into 49 graph nodes.

The Task 3 XAI notebook estimates graph-node importance using gradient-based node attribution. The resulting importance values are reshaped into a 7×7 representation so that the contribution of graph regions can be visualized spatially.

The node-importance analysis therefore directly explains the graph component of the final model rather than only the CNN feature extractor.

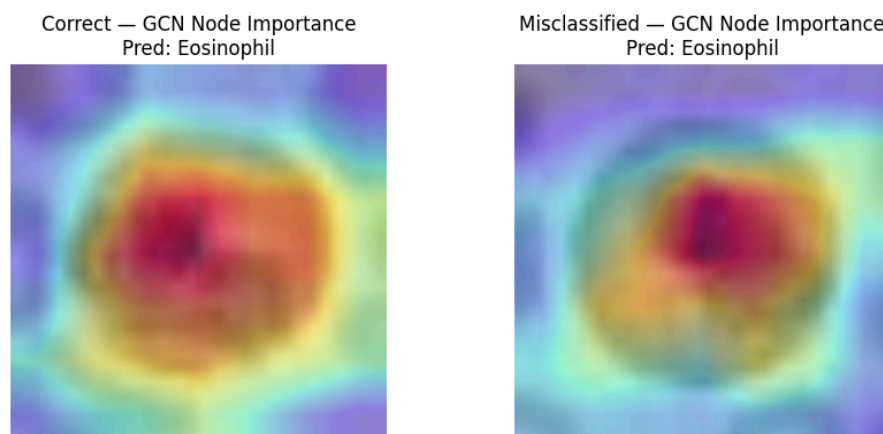


Figure 19. GCN node-importance maps for the correct and misclassified predictions.

19. Combined Explainability Analysis

A combined visualization was produced containing:

- original image;
- Grad-CAM;
- LIME;
- GCN node importance.

This allows the three explainability methods to be compared for the same correct and misclassified examples.

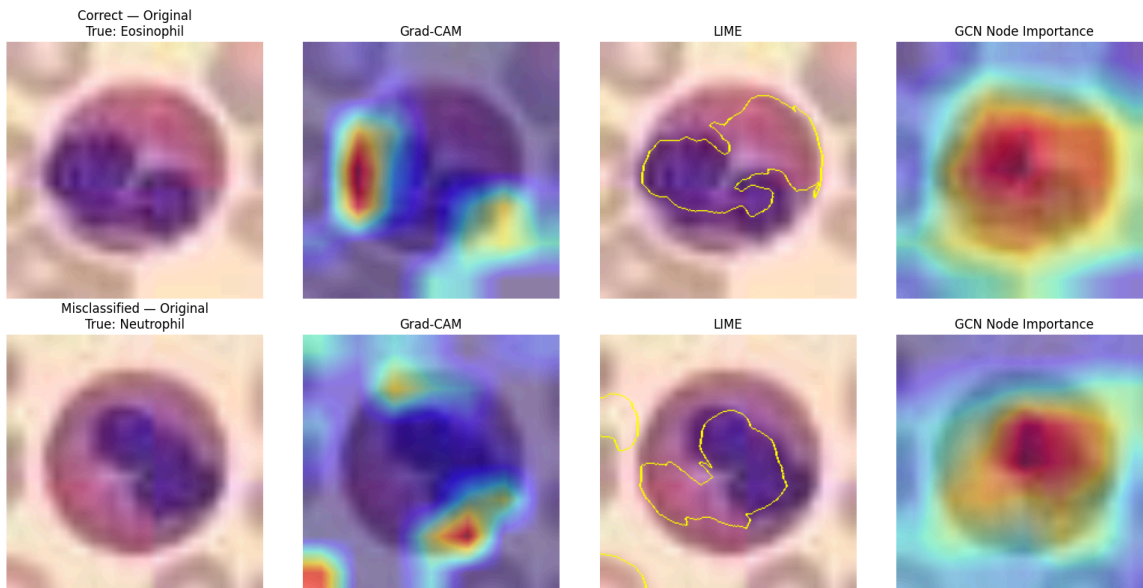


Figure 20. Combined explainability comparison using the original image, Grad-CAM, LIME and GCN node importance.

The three methods provide complementary forms of explanation. Grad-CAM highlights influential CNN feature-map regions, LIME provides perturbation-based local image explanations, and GCN node importance shows the estimated influence of the 49 graph nodes.

The notebook does not establish a specific biological interpretation of the highlighted regions. Therefore, the report does not claim that a particular nucleus, granule, or cellular structure was responsible unless it is directly visible and interpretable in the exported figures.

20. Overall Results

The major final results are summarized below.

Model	Accuracy	Macro-F1	Macro ROC-AUC
MobileNetV3Large — Task 2 reported	91.26%	90.38%	—
MobileNetV3Large — Task 3 reproduced	91.08%	90.28%	99.26%
Original ResNet50 + Attention + Residual GCN	91.61%	90.59%	99.23%
Final ResNet50 + Residual GCN	91.82%	91.18%	99.33%

The final GCN improved test accuracy and macro-F1 relative to both the reproduced CNN baseline and the reproduced original GCN.

The five-fold cross-validation results also favored the final GCN on macro-F1 in every fold:

- MobileNetV3Large: **89.34% $\pm$ 0.39%**
- Final Residual GCN: **90.62% $\pm$ 0.70%**
- Mean gain: **+1.28 percentage points**

However:

- McNemar exact p = **0.201397**
- Wilcoxon signed-rank p = **0.062500**

Therefore, the observed improvement should be described as **consistent and promising but not statistically significant at $\alpha = 0.05$** .

21. Discussion

The Task 3 experiments provide several important findings.

First, removing the GCN reduced validation macro-F1 from 89.97% to 89.36% in the direct A0 comparison. This supports the usefulness of graph-based reasoning when the CNN and attention representation are otherwise kept comparable.

Second, removing attention slightly increased validation macro-F1 in the A1 experiment. This indicates that the attention module used in Task 2 was not consistently helpful under the Task 3 pipeline.

Third, the isolated graph-connectivity experiment showed that the eight-neighbor graph achieved higher macro-F1 than the four-neighbor graph. However, the final consolidated pipeline retained the

four-neighbor graph. Therefore, graph connectivity should be described as an investigated design variable rather than claiming that the eight-neighbor graph became the final selected graph.

Fourth, the depth experiments showed that two layers produced the strongest result among the explicitly tested depths, while the final consolidated configuration used three layers. This again demonstrates that the final architecture was the locked configuration produced by the notebook's improvement workflow and should not be presented as proof that every individual ablation setting was globally optimal.

Fifth, dropout = 0.30 produced the strongest final consolidated validation macro-F1, and Residual GCN outperformed GraphSAGE in the direct variant comparison. These two findings provided clear support for the final graph-learning configuration.

Overall, Task 3 demonstrates that a carefully tuned graph-based model can improve the observed classification performance over a strong pretrained CNN baseline, while the ablation study also reveals that some components of the original Task 2 architecture were not necessary for the final model.

22. Limitations

Several limitations should be acknowledged.

First, BloodMNIST contains low-resolution 28×28 images. Although the images are resized to 224×224 for transfer learning, resizing does not create additional biological information.

Second, the graph nodes correspond to spatial locations in the learned 7×7 CNN feature map rather than explicitly segmented biological structures. The graph therefore represents learned spatial regions rather than manually annotated cellular parts.

Third, the statistical tests have limited power because the Wilcoxon analysis is based on only five paired folds.

Fourth, the Task 2 MobileNetV3Large result was not reproduced exactly in Task 3. The difference was small, but this demonstrates that training and evaluation of deep networks can vary across runs and environments.

Finally, the ablation study used a sequential experimental workflow rather than a complete factorial search over every combination of graph connectivity, GCN depth, dropout and architecture. The final configuration should therefore be interpreted as the locked configuration produced by the implemented workflow, not as proof of global architectural optimality.

23. Conclusion

Task 3 improved and evaluated the BloodMNIST graph-based classifier through controlled ablations, held-out test evaluation, five-fold cross-validation, significance testing and explainability analysis.

The final locked architecture consists of a frozen ImageNet-pretrained ResNet50 backbone, a 7×7 feature map represented by 49 graph nodes, a four-neighbor spatial graph, and a three-layer Residual GCN with hidden dimensions $256 \rightarrow 256 \rightarrow 128$ and GCN dropout of 0.30.

The final model achieved:

- **91.82% test accuracy**
- **91.74% macro precision**
- **90.87% macro recall**
- **91.18% macro-F1**
- **99.33% macro ROC-AUC**
- **0.9043 Cohen's Kappa**
- **0.9049 MCC**

Compared with the Task 2 reported MobileNetV3Large baseline, the final model improved accuracy by **0.56 percentage points** and macro-F1 by **0.80 percentage points**.

Five-fold cross-validation gave:

- MobileNetV3Large: **$89.34\% \pm 0.39\%$ macro-F1**
- Final Residual GCN: **$90.62\% \pm 0.70\%$ macro-F1**

The final GCN achieved higher macro-F1 in all five folds.

Nevertheless, the exact McNemar test produced $p = 0.201397$, and the two-sided Wilcoxon signed-rank test produced $p = 0.062500$. Neither result is statistically significant at $\alpha = 0.05$.

The explainability analysis further examined one correct Eosinophil prediction and one misclassified Neutrophil prediction using Grad-CAM, LIME and GCN node importance.

Overall, the Task 3 experiments show that the improved Residual GCN provides a modest and consistently observed performance advantage over the CNN baseline, while also demonstrating the importance of evaluating individual architectural components rather than assuming that every component improves performance.